

Internship in Physical Organic Chemistry

Structure Characterisation Sheet — Positive Ion Mode

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Fill in one copy of this sheet for the positive ion mode data. Work through the sections in order. Section 1 covers the MS1 spectrum; Section 2 covers the MS/MS fragment table; Section 3 is open space for the RDA argument and structure sketches.

Section 1 — MS1 spectrum (positive ion mode)

Compound: _____ Date: _____

Solvent: _____ Concentration: _____ Raw file: _____

List all peaks observed in the positive MS1 spectrum.

For each peak, record the observed m/z , the proposed ion identity, and whether the assignment is confirmed (by isotope simulation or mass accuracy) or tentative.

Observed m/z	Proposed identity	Status	Notes
_____	_____	☐ confirmed ☐ tentative	☐ _____
_____	_____	☐ confirmed ☐ tentative	☐ _____
_____	_____	☐ confirmed ☐ tentative	☐ _____
_____	_____	☐ confirmed ☐ tentative	☐ _____
_____	_____	☐ confirmed ☐ tentative	☐ _____
_____	_____	☐ confirmed ☐ tentative	☐ _____

Isotope simulation check (molecular ion):

Observed m/z : _____ Theoretical m/z : _____ Mass error (ppm): _____

Isotope pattern matches simulation: ☐ Yes ☐ No ☐ Partially

Precursor selected for MS/MS: _____ Reason: _____

Section 2 — MS/MS fragment table (positive ion mode)

Precursor ion (m/z): _____ NCE used for this table: _____ Raw file: _____

List all fragment ions observed at the selected NCE value.

Calculate the neutral loss for each fragment. Propose an assignment where possible. Mark RDA fragments with an asterisk (*) in the first column.

Fragment m/z	Neutral loss (Da)	Proposed formula of neutral	Proposed assignment / notes
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Effect of NCE on fragmentation pattern:

Compare the spectra at NCE low, medium, and high. Note which fragments appear only at higher NCE (higher-barrier pathways).

Section 3 — RDA argument and structure sketches (positive ion mode)

Use this space to work out the RDA fragmentation argument. Sketch the precursor structure, show the C-ring cleavage, and label the two resulting fragments. Indicate which fragment is observed as a charged ion and which is lost as a neutral.

RDA fragment observed at m/z : _____ Neutral loss from precursor:
_____ Da

Proposed identity of charged fragment: _____

Proposed identity of neutral fragment (lost): _____

Structure sketch: (draw here)

Is this assignment consistent with the Anchor 2 paper?

☐ Yes ☐ No ☐ Partially ☐ Not yet checked

Notes: _____
